Active Components

Component	Selected Model	Link	Notes
Flight Controller & ESC	SPEEDYBEE F405 V4 & 60A ESC	link	This is so far the best model to choose from due to its high quality and the availability of an instruction manual.
MCU + Camera	ESP32-CAM version Upgraded ESP32 camera component (OPTIONAL)	Link Link	This MCU offers enough performance speed to process the communication protocols. Plus, this has a built-in camera, which makes it an ideal component for our project. Although it takes low-quality pictures therefore we provided an extra link to upgrade the ESP32-Camera simply by removing the existing one to an upgraded version.
Battery (Li-ion or Li-Po)	ZEEE 3S (11.1V), 50C Rating, 5200mAh capacity,	link	VERY IMPORTANT TO SELECT THE FOLLOWING SPECS: 1- 3S (11.1V) or more. 2- at least 5000 mAh capacity, 3- C rating at least 50C. 4- XT60 CONNECTOR!!, with voltage balance wire connection included for the charging. 5- size that can fit the frame. 6- Good brands such as ZEEE, since many of the options are not high-quality.
BLDC Motors	E-power 4pcs X3115 3115 900KV 3-6S Brushless Motor	link	The frame only supports the motors of the following specs: 1- 900 KV. 2- 3115 dimension specs.

Passive Components

Component	Selected Model	Link	Notes
Drone's Frame	MAK4 V2 10-inch	link	The frame must be: 1- carbon-fiber. 2- Support landing gear if possible. 3- not tiny nor too big, 10-inch is good.
Drone's Propeller	2Pairs(2CW+2CCW) SoloGood 1050 10X5X3 3-Blade Glass Fiber Nylon Propeller for Multirotor 10" FPV Cinelifter MarcoQuad	link	The frame only supports the following props specs; 1- 1050 (10x5). 2- 3 blades if possible. 3-2Pairs(2CW+2CCW)

Soldering Components (IMPORTANT!)

Component	Selected Model	Link	Notes
Soldering Iron		link	This kit is sufficient for the project, but do not use the SOLDERING WIRE! It will damage your board.
Soldering Iron Tip		link	Highly important to use only the K type!. Using something else will not going to ease your task.
Soldering wire		link	It's very important to get the 99.3% Sn option with FLUXx; anything besides that will be useless for the project and will DAMAGE YOUR BOARD.
Tip Cleaner		Link	Using the soldering tip over and over will reduce its performance, using this item is very important to keep the tip cleaned.
Soldering wick		link	Very useful for desoldering, since sometimes using the soldering pump suction is not enough.

Highly Important Tools

Component	Selected Model	Link	Notes
Wire Tie		link	
Wire stripper		link	
Battery charger		link	
Jumper wires (all types)		link	
Breadboard		Link	For running tests.
Multimeter		Link	For running tests, especially to test your soldering and ensure that the terminals are not connected unintentionally.
Motor Screws		link	Using the screws provided by the motors is not the right choice because they have a coating that is susceptible to scratching and crumbling, which sticks to the motors and disrupts the motors' speed. Make sure to get the following screws specs: 1- Self-Locking Nuts. 2- Nylon Lock Nut Set 3- M5 4- hex type 5- good quality, no color paint
BLDC Motors Screw Driver		link	Very crucial to have this, DO NOT INSTALL THE MOTOR SCREWS USING ANY OTHER TOOLS. Its more likely to damage the screws and the motor all together.
ESC soldering test	ShortSaver 2 V2	Link	YOU NEED TO TEST YOUR ESC CONNECTION BEFORE PLUGGING IT DIRECTLY TO THE BATTERY. USE THIS TOOL BEFORE PROCEEDING TO THE MOTORS TEST.
Screwdrivers		Link	Must have magnetic tips